

January 2026 CSE 220
Online on Signals and Their Properties
Subsections: C1, C2
Total Marks: 10
Total Time: 40 Minutes

July 10, 2026

Any signal $x(t)$ can be decomposed into an even part and an odd part: $x(t) = x_e(t) + x_o(t)$.

In this assignment, you are given the even part $x_e(t)$ of a sampled signal in the provided Python template file (choose from 1-5 after running the file). Your task is to construct an odd signal $x_o(t)$ such that the final signal $x(t)$ is constant on the non-negative side. That is, for all sampled points with $t \geq 0$,

$$x(t) = M, \text{ for some constant value } M$$

In other words, the right side of the signal (including $t = 0$) must look like a constant horizontal line. Since every odd signal satisfies $x_o(0) = 0$, we must have $M = x_e(0)$.

You will compute $x_o(t)$ and plot $x_e(t)$, $x_o(t)$, and $x(t)$.

Sketching an example case, thinking a little, and doing simple mathematics will be sufficient to solve the problem.

Please first run the template code as given. Then write the appropriate code in the blocks marked "TODO". The code should work correctly for all five test cases, regardless of which even-signal option (1-5) is selected.

Tasks

1. Determine the value of M from the condition

$$M = x_e(0).$$

2. Construct the required odd component $x_o(t)$.
3. Plot $x_e(t)$, $x_o(t)$, and $x(t)$ in three separate subplots with proper titles.

Marks Distribution

- Determining M : 2 Marks
- Construction of the odd component $x_o(t)$: 4 Marks
- Plotting of $x_e(t)$, $x_o(t)$, and $x(t)$: 4 Marks